

Admin Role

Admin gets full oversight and override power across all 10 modules, plus some system-level capabilities that sit outside the module list entirely. Here's the breakdown:

1. Equipment Management

- Add, edit, deactivate, or delete equipment records (name, category, quantity, rental rate, condition)
- View the full inventory across all categories
- Override equipment status manually (e.g., force-flag as "under repair" or "retired")

2. Customer Management

- View, add, edit, or deactivate any customer account
- Reset customer credentials / unlock locked accounts
- View a customer's full history across reservations, transactions, and CRM notes

3. Reservation Management

- View all reservations system-wide (not just their own)
- Create, edit, or cancel any reservation — including overriding one Staff made
- Resolve double-booking conflicts manually

4. Rental Transactions

- View every transaction across the system
- Void or correct erroneous transactions
- Approve refunds or manual adjustments

5. Equipment Availability

- View the full availability calendar across all equipment
- Manually block or override availability (e.g., hold equipment for an internal reason)

6. Maintenance Management

- View all maintenance logs and schedules
- Approve, reassign, or close out maintenance tasks
- Set maintenance policies (e.g., recurring service intervals per equipment type)

7. Billing

- View all invoices and payment records system-wide
- Edit, waive, or correct billing errors and apply discounts
- Generate billing summaries across date ranges or customers

8. Customer CRM

- View and manage customer communication logs, notes, and complaints
- Configure CRM behavior (e.g., automated reminders, follow-up rules)

9. Damage & Penalty Tracking

- View all damage reports and penalty charges
- Approve, waive, or adjust penalty amounts
- Set the penalty rules/fee schedule used system-wide

10. Reports

- Generate and export all reports (revenue, equipment utilization, overdue rentals, customer activity, maintenance costs)
- View system-wide dashboards

System-level (Admin-only, outside the 10 named modules)

- Manage user accounts and role assignments — create Staff/Customer accounts, assign or revoke roles
- Configure system-wide settings (rental policies, tax rates, business hours)
- View system/audit logs (login history, action history, error logs)

Staff Role

Staff handles the day-to-day operational work — everything hands-on, nothing system-level. This role absorbed the former Rental Staff/Clerk and Inventory & Maintenance Officer duties, so it covers both the front-counter/reservation side and the equipment/maintenance side.

1. Equipment Management

- Add new equipment records and edit existing ones (rate, quantity, condition, description)
- Update equipment status (available, rented out, under repair)
- View full inventory list
- *Cannot* permanently delete/retire equipment — that's Admin-only

2. Customer Management

- Register new customer accounts (walk-in/phone bookings)
- Edit customer contact info and details
- View a customer's rental history
- *Cannot* deactivate accounts or reset login credentials system-wide — Admin-only

3. Reservation Management

- Create, edit, and cancel reservations
- Check availability before confirming a booking
- Resolve simple scheduling conflicts (suggest alternate dates/equipment)

4. Rental Transactions

- Record new transactions (rent-out and return)
- Process payments and generate receipts
- Update transaction status (paid, pending, completed)
- *Cannot* void transactions or approve refunds — flagged to Admin instead

5. Equipment Availability

- View the real-time availability calendar
- Update availability when equipment is rented out, returned, or pulled for maintenance

6. Maintenance Management

- Log maintenance needs and schedule service dates
- Update maintenance status (pending, in progress, completed)
- Record maintenance notes and costs

7. Billing

- Generate invoices tied to transactions
- Record payments received
- View a customer's billing history
- *Cannot* waive fees, apply discounts, or edit billing rules — Admin-only; discrepancies get flagged upward

8. Customer CRM

- Log customer communications (calls, inquiries, complaints)
- Add notes to a customer's profile
- Send standard follow-ups/reminders (using templates Admin configured)
- *Cannot* create new automation rules — Admin-only

9. Damage & Penalty Tracking

- Record damage reports on equipment return (notes, condition, photos if applicable)
- Apply standard penalty fees based on the fee schedule Admin set
- Flag damage beyond the standard fee schedule for Admin review/approval

10. Reports

- Generate operational reports relevant to their work (daily rental summary, overdue equipment, maintenance schedule)
- *Cannot* access full financial exports or system-wide dashboards — Admin-only

Boundary line vs. Admin: Staff has zero access to user/role management, system settings, audit logs, or final approval authority on refunds/penalty waivers. Every "cannot" above is deliberately routed back to Admin — that's what keeps the RBAC story clean for grading.

Customer Role

Customer is the self-service, external-facing role — view-only or self-scoped everywhere, and locked out entirely from anything internal/operational.

1. Equipment Management

- Browse the equipment catalog (name, category, photos if available, rate, condition)
- *Read-only* — cannot add, edit, or see internal notes on equipment

2. Customer Management

- View and edit their **own** profile (contact info, password)
- *Cannot* view or touch any other customer's account

3. Reservation Management

- Create a reservation request for available equipment/dates
- View their own upcoming and past reservations
- Cancel or modify their own *upcoming* reservation (within whatever cancellation window you define)
- *Cannot* see or touch anyone else's reservations

4. Rental Transactions

- View their own transaction history and receipts
- *Cannot* create or modify a transaction directly — that stays with Staff at pickup/return, since it involves physically verifying equipment

5. Equipment Availability

- View real-time availability while browsing/booking
- *Read-only* — no edit access

6. Maintenance Management

- **No access.** This is a purely internal module; customers have no reason to see maintenance logs

7. Billing

- View their own invoices and payment status
- Pay online, if that's a feature you build in
- *Cannot* view other customers' billing or edit any billing record

8. Customer CRM

- Submit an inquiry, request, or complaint (this feeds into Staff's CRM view)
- *Cannot* see internal notes Staff/Admin have logged about them — those stay internal

9. Damage & Penalty Tracking

- View damage reports and penalty charges applied to their *own* account, with the reason/amount
- *Cannot* edit or waive anything — can only view, and optionally flag a dispute that routes to Staff/Admin

10. Reports

- **No access.** Reports are an internal management tool

System-level (Customer-only, outside the 10 modules)

- Self-registration (create their own account)
- Manage notification preferences, if you build that in

Boundary line vs. Staff/Admin: Customer never sees anyone else's data, never touches internal operational tools (maintenance, reports, CRM notes, penalty rules), and every write action is scoped to "their own record only." That's the cleanest possible RBAC boundary for grading — it's obvious at a glance why Customer is the most restricted role.